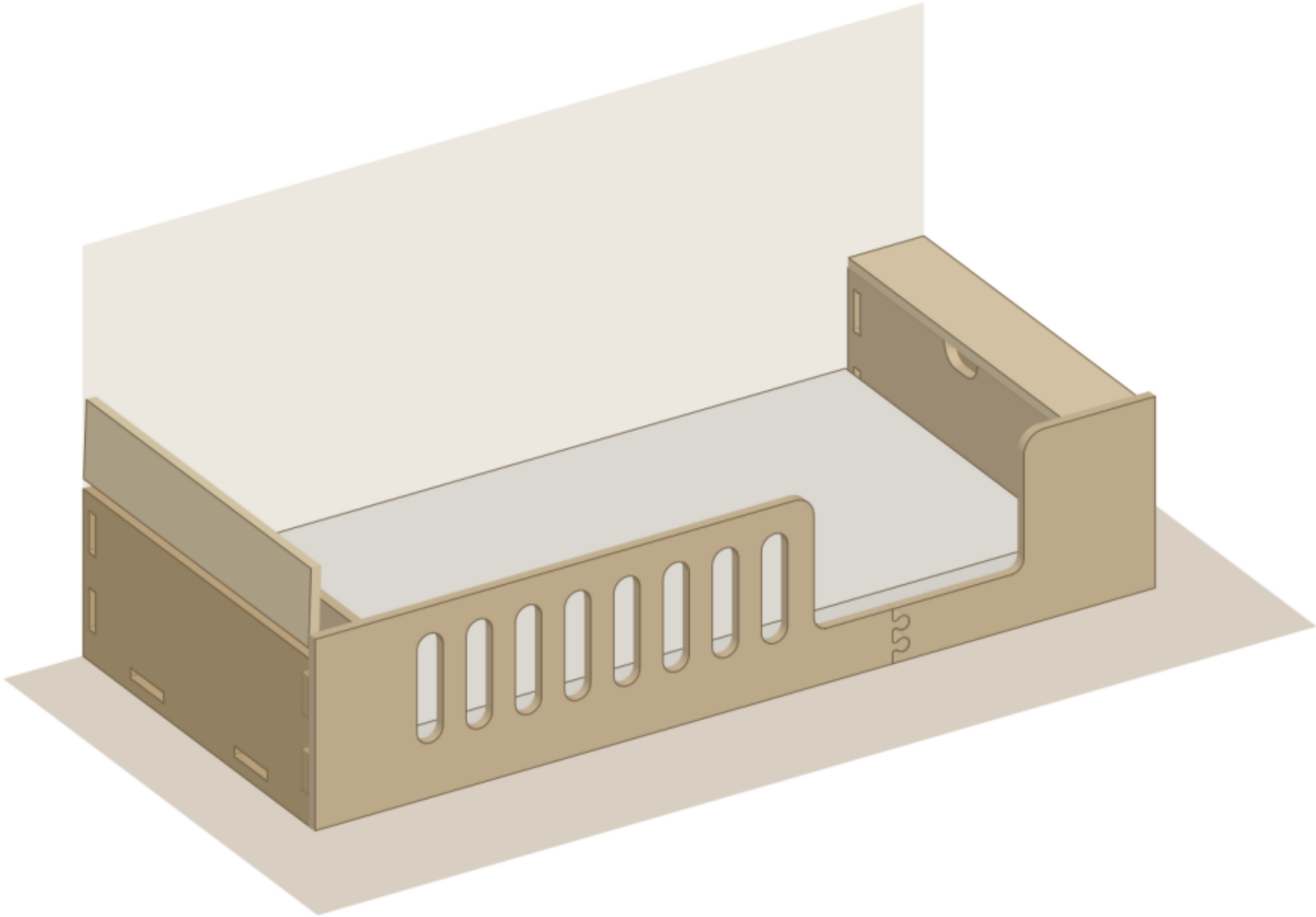
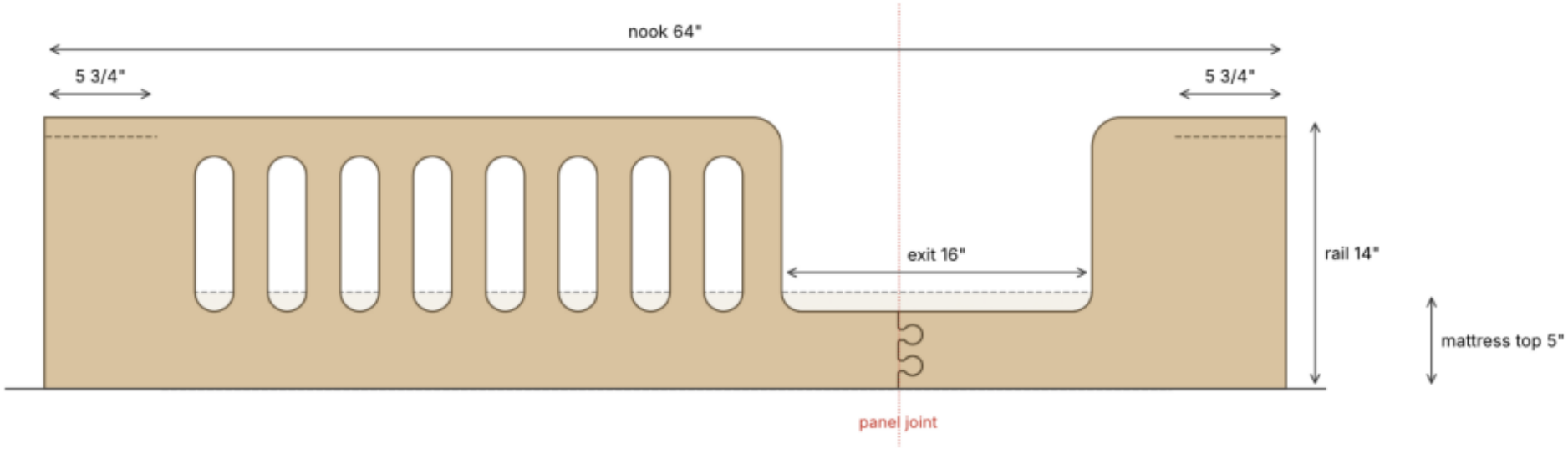
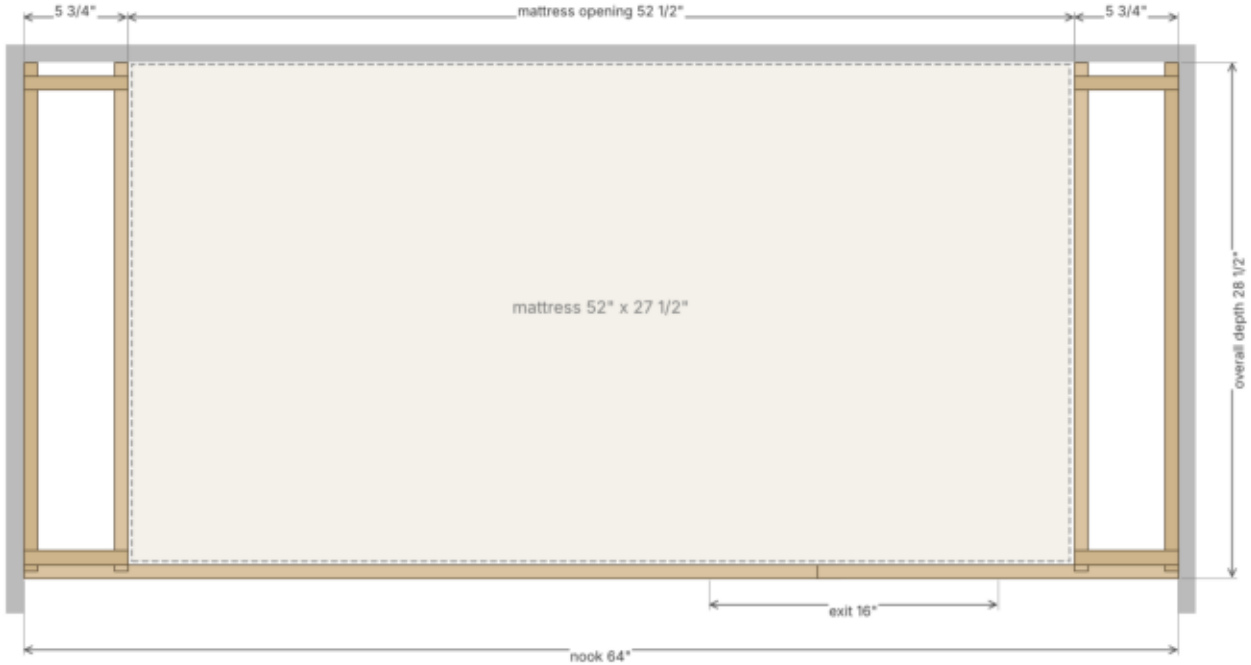
Floor bed, side exit



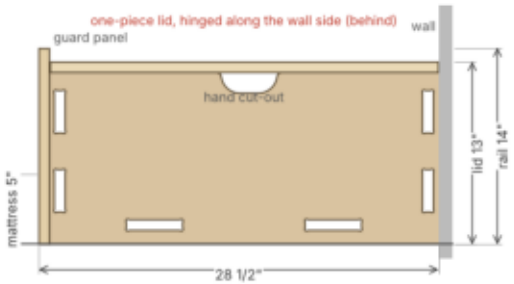
Floor bed, side exit: front view



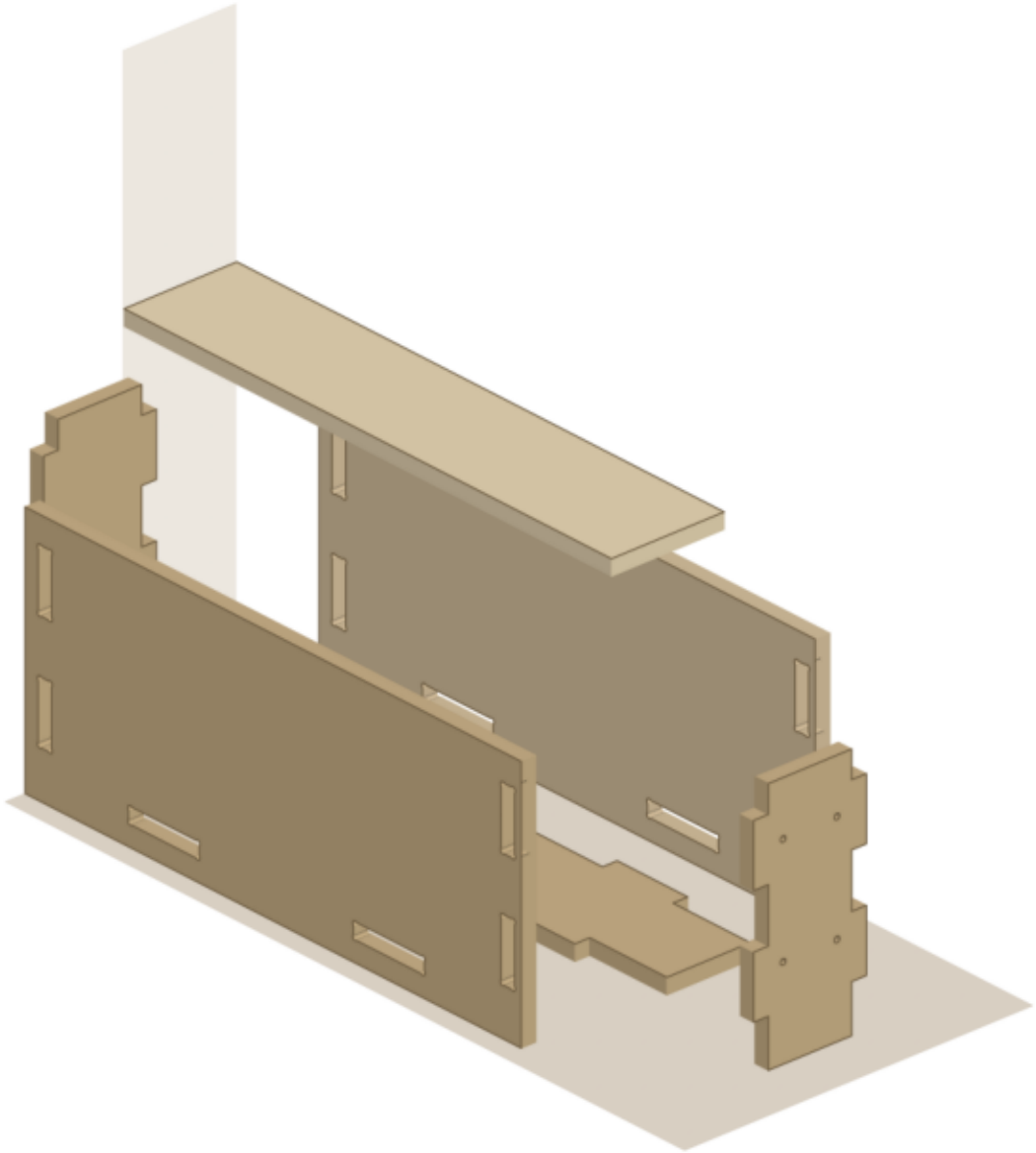
Plan (front of bed at the bottom)



Box seen from the mattress

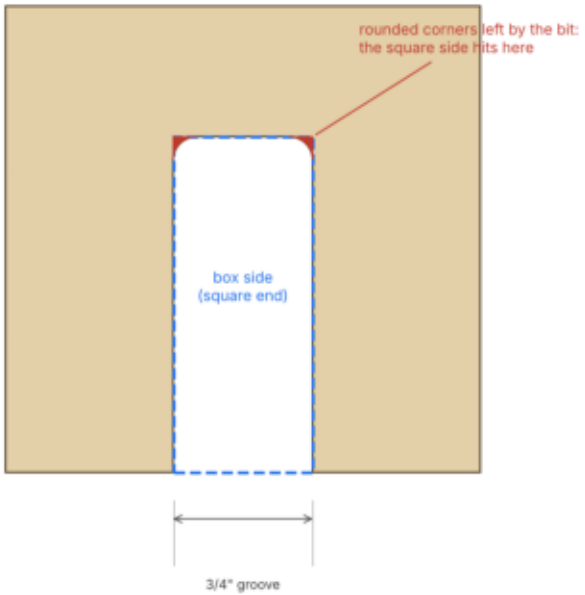


Head box, exploded (tabs pass through the sides)

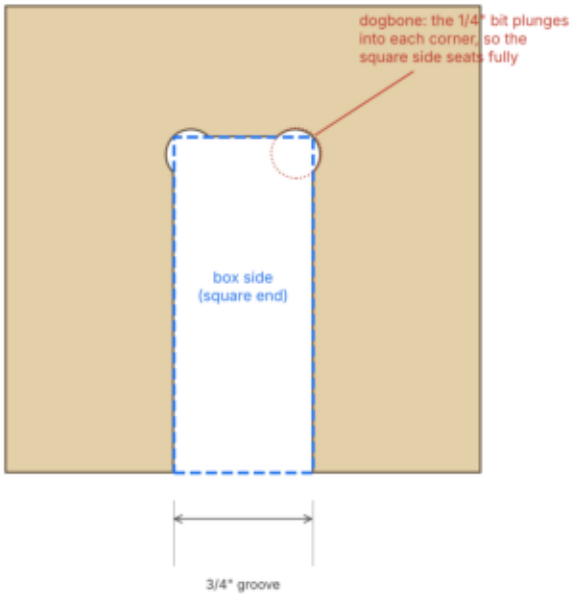


Dogbones: how a square part fits a CNC-cut groove (looking at the back of the front panel)

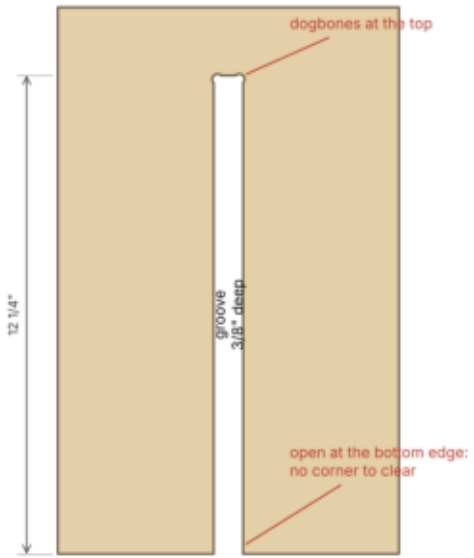
Without dogbones (red = wood that blocks the square corner)



With dogbones

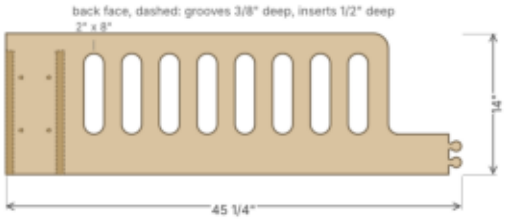


Back of the front panel, one groove

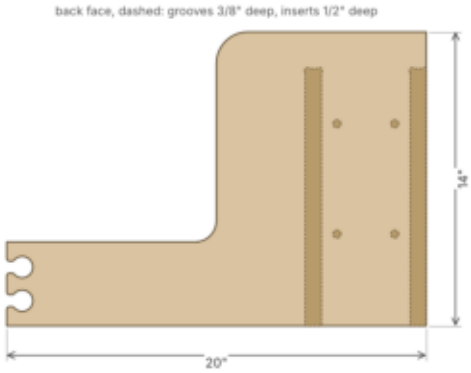


Floor bed, side exit: part drawings (inches, 3/4" plywood)

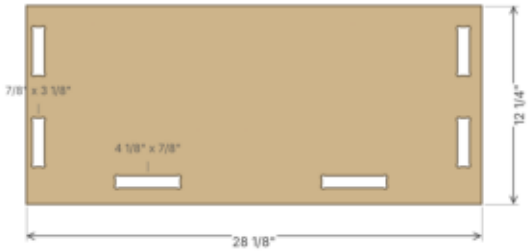
Front panel A (qty 1)
knob half



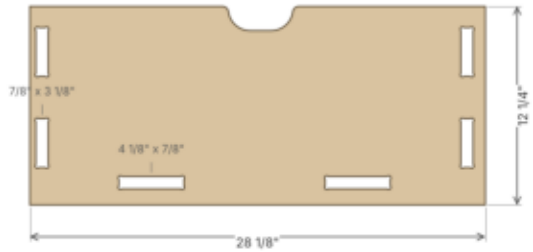
Front panel B (qty 1)
socket half



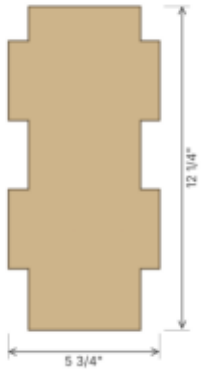
Box side (wall) (qty 2)
against the nook wall, 1 per box



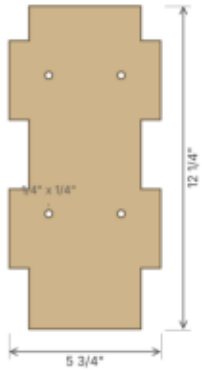
Box side (mattress) (qty 2)
faces the mattress; hand cut-out to open the lid



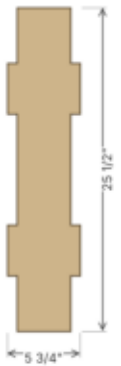
Box back (qty 2)
screw through into wall studs



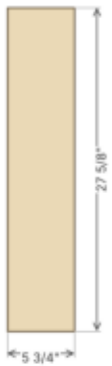
Box front (qty 2)
against the back of the front panel; 4 bolts into it



Box floor (qty 2)
sits 1" off the floor

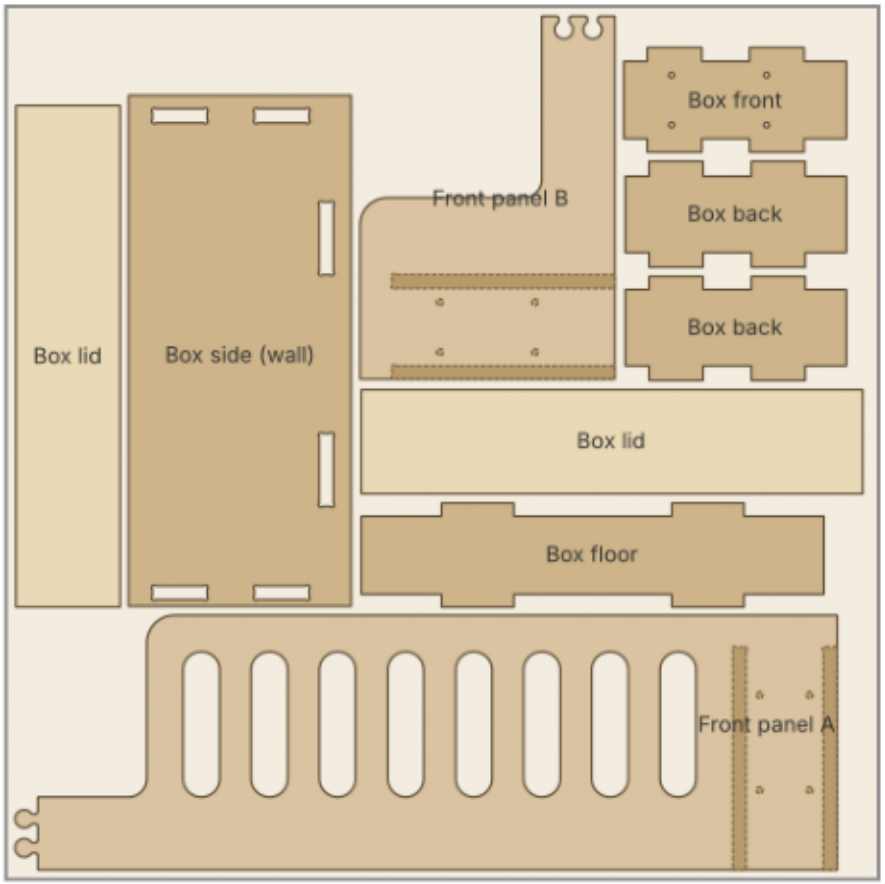


Box lid (qty 2)
piano hinge along the long wall-side edge

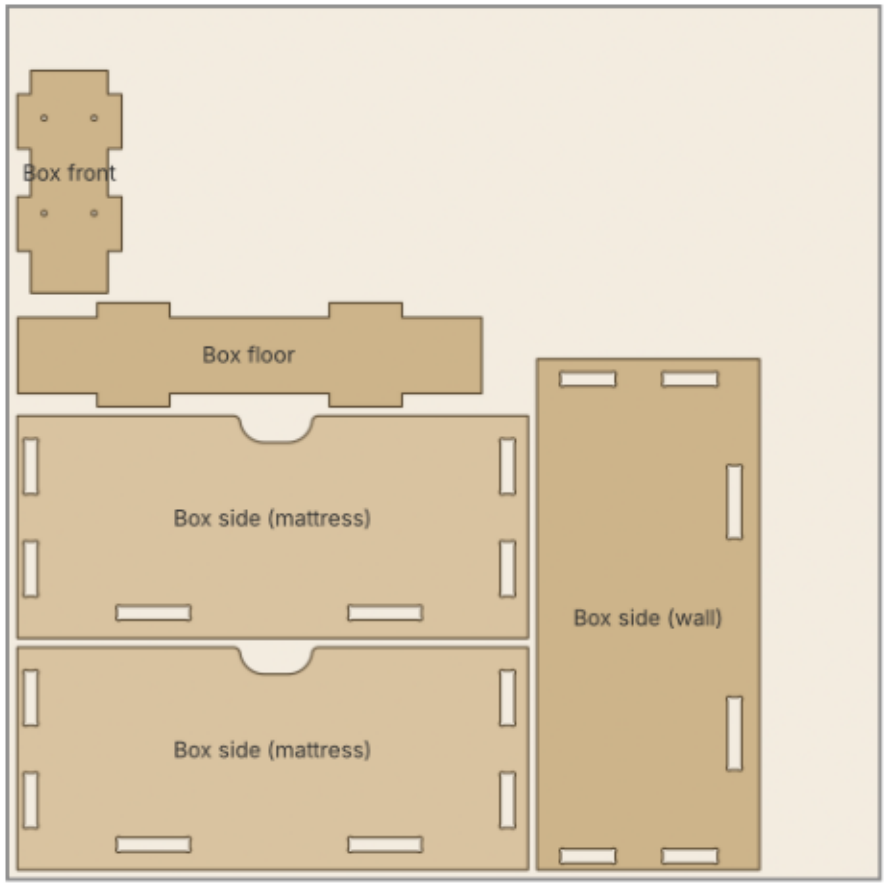


Floor bed, side exit: nesting on 48" square panels

4'x4' panel 1



4'x4' panel 2



Floor bed, side exit: cut list, hardware, assembly

Cut list

Part	Qty	Size (W x H)	Notes
Front panel A	1	45 1/4" x 14"	knob half
Front panel B	1	20" x 14"	socket half
Box side (wall)	2	28 1/8" x 12 1/4"	against the nook wall, 1 per box
Box side (mattress)	2	28 1/8" x 12 1/4"	faces the mattress; hand cut-out to open the lid
Box back	2	5 3/4" x 12 1/4"	screw through into wall studs
Box front	2	5 3/4" x 12 1/4"	against the back of the front panel; 4 bolts into it
Box floor	2	5 3/4" x 25 1/2"	sits 1" off the floor
Box lid	2	5 3/4" x 27 5/8"	piano hinge along the long wall-side edge

Key dimensions

Panels: 2 pieces of 4'x4' (1 x 4'x8' sheet(s) of 3/4" plywood, each cut in half)
Heights from floor: mattress top 5", rail top 14", box lids 13", exit sill 4"
Exit opening: 38" to 54" from the head wall; panel halves join at 44"
Head/foot boxes: 5 3/4" wide outside, 4 1/4" inside, 27 3/4" deep

Hardware

- Wood glue (PVA)
- 1-1/4" wood screws x 24 (box joinery)
- 1/4-20 threaded inserts for wood x 8, no longer than 3/8" (front panel)
- 1/4-20 x 1" bolts + washers x 8 (box fronts to front panel)
- 2-1/2" screws x 4 into studs (box backs)
- Piano hinge 1-1/2" open width, 2 x 24" length(s) (1 per lid)
- Soft-close lid stays x 2 (light-duty toy-box type)
- Shims as needed

Assembly

1. Cut all parts (sheets/panel_*.dxf). The two front-panel pieces are mirrored in the files: cut them BACK FACE UP so the POCKET layers (box-side grooves 3/8" deep, insert holes 1/2" deep) land on the back. Profile CUT_INSIDE before CUT_OUTSIDE, use tabs/onion skin on small parts, and dry-fit the puzzle joint and one groove on scrap first; adjust knob_clearance / tab_clearance if needed.
2. Ease every exposed edge with a 1/4" roundover (rail top, slots, exit opening, lid edges and the hand cut-outs). Leave joinery edges and faces that meet the wall square.
3. Boxes: glue the front, back and floor tabs into one side, add the second side, clamp square. Tabs are flush with the outside faces. Pre-drill and screw (1-1/4") through the side into each